

Restaurant Food Demand Prediction and Inventory Planning Using Machine Learning

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Section: BCADS33

Course: BCA (Data Science & Artificial Intelligence)

Academic Year: 2026–27

Technology: Python, Machine Learning, Jupyter Notebook

1. Abstract

Food demand prediction is an important task in the food-service industry because businesses need to prepare the right quantity of food according to expected customer demand. Overestimating demand can result in food wastage and unnecessary operational costs, while underestimating demand can cause stock shortages and customer dissatisfaction.

This project develops a machine learning-based food demand prediction system using historical food order data. The dataset contains information related to weekly food orders, meals, fulfillment centers, prices, promotional activities, meal categories, cuisines, and operational areas.

The project follows a complete machine learning workflow including data preprocessing, data integration, exploratory data analysis, feature engineering, categorical encoding, time-based train-validation splitting, model training, and model evaluation.

Three regression algorithms were implemented and compared: Linear Regression, Random Forest Regressor, and XGBoost Regressor. The models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

Among the evaluated models, Random Forest achieved the best overall performance with an MAE of **88.32**, RMSE of **196.48**, and R^2 Score of **0.714**. The final Random Forest model was trained using the complete historical dataset and used to generate future food demand predictions.

The predicted demand can be used for food preparation planning, inventory management, reduction of food wastage, and prevention of stock shortages.

2. Introduction

The food-service industry deals with continuously changing customer demand. The number of food orders can vary according to factors such as week, meal type, price, promotional activities, and location of the fulfillment center.

Accurate demand forecasting allows food-service businesses to make better operational decisions. If a business prepares significantly more food than required, unused food may become waste. On the other hand, preparing less food than the actual demand may result in unavailable meals and lost sales.

Machine learning provides an effective approach for identifying patterns in historical data and using those patterns to estimate future demand.

In this project, historical food order data is analysed to develop a machine learning model capable of predicting the number of orders. The project also demonstrates how predicted demand can be converted into a practical food preparation recommendation.

3. Problem Statement

Food-service businesses need to estimate future food demand accurately to manage food preparation and inventory efficiently.

Traditional estimation methods may not adequately capture the relationship between demand, pricing, promotions, meal characteristics, and fulfillment-center information.

Therefore, the problem addressed in this project is:

"To develop a machine learning-based system that predicts food demand using historical order patterns, meal information, pricing, promotional activities, and fulfillment-center characteristics."

The predicted demand can then be used to support food preparation and inventory planning decisions.

4. Objectives

The major objectives of this project are:

1. To analyse historical food order data.
2. To understand the factors affecting food demand.
3. To perform data preprocessing and data integration.
4. To perform exploratory data analysis using visualizations.
5. To create useful features for machine learning.

6. To encode categorical information into numerical form.
7. To develop multiple regression models for food demand prediction.
8. To compare Linear Regression, Random Forest, and XGBoost.
9. To evaluate models using MAE, RMSE, and R² Score.
10. To select the best-performing model.
11. To predict future food demand.
12. To provide a preparation quantity recommendation based on predicted demand.
13. To demonstrate the practical application of machine learning in food-service operations.

5. Dataset Description

The project uses a food demand forecasting dataset containing historical food order information.

The dataset consists of four major files:

5.1 train.csv

This file contains historical food order records and the target variable `num_orders`.

Important columns include:

Column	Description
id	Unique record identifier
week	Week number
center_id	Fulfillment center identifier
meal_id	Meal identifier
checkout_price	Final selling price
base_price	Original/base price
emailer_for_promotion	Indicates email promotion
homepage_featured	Indicates homepage promotion
num_orders	Number of orders / target variable

5.2 meal_info.csv

This file contains information about individual meals.

Important attributes include:

- meal_id
- category
- cuisine

5.3 fulfilment_center_info.csv

This file contains information about fulfillment centers.

Important attributes include:

- center_id
- city_code
- region_code
- center_type
- op_area

5.4 test_QoiMO9B.csv

This file contains records for which the model generates future demand predictions.

The target variable `num_orders` is not available for these records.

6. Technologies Used

The following technologies and libraries were used:

Programming Language

Python

Python was selected because of its extensive data science and machine learning ecosystem.

Development Environment

Jupyter Notebook

Jupyter Notebook was used for data analysis, visualization, model development, and experimentation.

Libraries

Pandas: Used for data loading, manipulation, merging, and analysis.

NumPy: Used for numerical operations and mathematical transformations.

Matplotlib: Used for creating visualizations.

Seaborn: Used for statistical visualizations and graphs.

Scikit-learn: Used for preprocessing, model training, and evaluation.

XGBoost: Used to implement the XGBoost regression model.

7. Methodology

The project follows the following workflow:

Historical Food Demand Data

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Data Collection

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Data Loading

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Data Understanding

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Data Integration

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Exploratory Data Analysis

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Feature Engineering

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Categorical Encoding



Time-Based Train/Validation Split



Machine Learning Model Training



Model Evaluation



Best Model Selection



Future Demand Prediction



Inventory & Food Preparation Recommendation

8. Data Preprocessing and Integration

The dataset was first loaded into Python using Pandas.

The historical training data was combined with meal information using `meal_id` and fulfillment-center information using `center_id`.

The merging process allowed the model to use additional information such as:

- Food category
- Cuisine
- City
- Region
- Center type
- Operational area

The merged dataset provided a more comprehensive representation of the factors influencing food demand.

Missing values and duplicate records were also checked during the preprocessing stage.

9. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the distribution and relationships within the dataset.

Several analyses were conducted.

9.1 Distribution of Food Orders

The distribution of `num_orders` was analysed to understand the range and frequency of food orders.

This analysis helped identify the general demand pattern and the presence of highly demanded observations.

9.2 Weekly Demand Analysis

Total food orders were grouped according to the week number.

The weekly demand graph helped identify changes in food demand over time and provided an understanding of demand trends.

9.3 Promotion Analysis

The average number of orders was compared between promoted and non-promoted meals.

This analysis helped understand whether promotional activities were associated with differences in demand.

9.4 Homepage Featured Analysis

The average number of orders was compared between meals featured on the homepage and meals that were not featured.

9.5 Cuisine Analysis

Average demand was calculated for different cuisines.

This helped identify differences in demand across cuisine types.

9.6 Category Analysis

Average orders were calculated for different food categories.

This provided insights into which types of meals were associated with higher demand.

9.7 Price Analysis

The relationship between checkout price and number of orders was visualized using scatter plots.

This helped analyse the relationship between pricing and demand.

10. Feature Engineering

Feature engineering was performed to create additional variables that could provide useful information to the machine learning models.

10.1 Discount

The discount amount was calculated as:

$$\text{Discount} = \text{Base Price} - \text{Checkout Price}$$

10.2 Discount Percentage

The percentage discount was calculated as:

$$\text{Discount Percentage} = ((\text{Base Price} - \text{Checkout Price}) / \text{Base Price}) \times 100$$

10.3 Price Ratio

The price ratio was calculated as:

$$\text{Price Ratio} = \text{Checkout Price} / \text{Base Price}$$

These features helped represent the pricing relationship more clearly.

11. Categorical Encoding

Machine learning algorithms generally require numerical input.

Therefore, categorical variables such as:

- Category
- Cuisine
- Center Type

were converted into numerical representations using **Label Encoding**. The encoded variables were then included in the model features.

12. Train-Validation Split

Because the project focuses on demand forecasting, a time-based validation approach was used instead of a completely random split.

Earlier weeks were used for model training, while later weeks were used for validation.

This approach better represents the real-world forecasting situation where historical information is used to predict future demand.

The target variable was also log-transformed using:

$\log(1 + \text{num_orders})$

This transformation helps reduce the effect of extreme values and skewness in the demand distribution.

13. Machine Learning Models

Three regression algorithms were implemented.

13.1 Linear Regression

Linear Regression was used as a baseline model.

It attempts to model the relationship between input features and the target variable using a linear equation.

It provides a simple benchmark against which more complex models can be compared.

13.2 Random Forest Regressor

Random Forest is an ensemble machine learning algorithm that combines multiple decision trees.

Each tree learns different patterns from the data, and their predictions are combined to produce the final prediction.

Random Forest can capture nonlinear relationships between features and demand.

In this project, Random Forest produced the best overall performance.

13.3 XGBoost Regressor

XGBoost is an advanced gradient boosting algorithm.

It builds decision trees sequentially, with each new tree attempting to improve the errors made by previous trees.

XGBoost is widely used for structured/tabular datasets and can capture complex relationships between variables.

14. Model Evaluation

The models were evaluated using three metrics.

14.1 Mean Absolute Error (MAE)

MAE represents the average absolute difference between actual and predicted values.

Lower MAE indicates better performance.

14.2 Root Mean Squared Error (RMSE)

RMSE measures the square root of the average squared prediction error.

RMSE gives greater importance to larger errors.

Lower RMSE indicates better performance.

14.3 R² Score

R² indicates how much variation in the target variable is explained by the model.

A higher R² generally indicates better model performance.

15. Model Comparison

The obtained results were:

Model	MAE	RMSE	R ²
Linear Regression	164.56	327.34	0.205
Random Forest	88.32	196.48	0.714
XGBoost	86.93	203.52	0.693

Result Analysis

Linear Regression produced the weakest overall performance.

XGBoost achieved a slightly lower MAE than Random Forest, but its RMSE was higher and its R² score was lower.

Random Forest achieved:

- MAE = **88.32**
- RMSE = **196.48**
- R² = **0.714**

Therefore, **Random Forest Regressor was selected as the final model** based on its overall performance across the evaluation metrics.

16. Feature Importance

Feature importance was calculated using the Random Forest model.

Feature importance indicates how useful different input variables were in making predictions.

The feature-importance graph generated in the notebook can be included in this section.

Interpretation

The feature importance analysis helps identify which variables contribute most strongly to demand prediction.

This information can help businesses understand which operational factors should receive greater attention when planning food demand.

Note: In the final report, mention the actual top features according to your generated feature-importance graph rather than assuming them in advance.

17. Future Demand Prediction

After selecting Random Forest as the best model, the model was retrained using the complete historical dataset.

The trained model was then applied to the test dataset.

The model generated predicted order quantities for future records.

A prediction file named:

food_demand_predictions.csv

was created containing:

- **id**
- **Predicted_orders**

Example predictions generated by the model include:

ID	Predicted Orders
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1028232	99
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1127204	81
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1212707	133
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1082698	58
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1400926	59
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1284113	206
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1197966	277
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These predictions demonstrate how the trained model can estimate future food demand.

18. Inventory and Food Preparation Recommendation

The predicted demand can be converted into a practical preparation recommendation.

In this project, a **5% preparation buffer** was demonstrated.

The recommendation was calculated as:

Recommended Preparation = Predicted Orders $\times$ 1.05

For example, if predicted demand is 200 orders:

Recommended Preparation = 200 $\times$ 1.05 = 210

The buffer is intended to provide additional capacity for unexpected demand.

In a real production system, the buffer should be determined using historical forecast errors, service-level requirements, perishability, and business costs rather than using a fixed percentage for every situation.

19. Business Applications

The developed system can support several business activities.

19.1 Food Preparation Planning

Predicted demand can help determine the quantity of food that should be prepared.

19.2 Inventory Management

Demand forecasts can help businesses plan ingredient and meal inventory.

19.3 Food Waste Reduction

Better demand estimates can reduce excessive food preparation.

19.4 Stock Shortage Prevention

Demand forecasting can help businesses prepare sufficient quantities during high-demand periods.

19.5 Promotional Planning

Demand patterns can help businesses understand the relationship between promotional activities and orders.

19.6 Operational Decision-Making Managers can use demand predictions to support staffing, preparation, and supply planning decisions.

20. Limitations

The project has several limitations:

1. The dataset does not contain detailed customer-level information.
2. Weather conditions are not included.
3. Holiday and festival information is not included.
4. Unexpected events may cause sudden changes in demand.
5. Real-time restaurant/POS integration has not been implemented.
6. The model is primarily based on historical patterns.

7. A fixed preparation buffer was used only as a demonstration.

21. Future Scope

The project can be further improved by:

1. Integrating real-time restaurant order data.
2. Adding weather information.
3. Including holidays and festival calendars.
4. Incorporating local events and seasonal information.
5. Using advanced time-series forecasting techniques.
6. Developing a web-based prediction dashboard.
7. Creating real-time demand alerts.
8. Integrating the prediction system with inventory management software.
9. Automatically adjusting preparation quantities according to forecast uncertainty.
10. Deploying the model as an API or web application.

22. Conclusion

This project successfully developed a machine learning-based food demand prediction system using historical food order data.

The project involved data integration, exploratory data analysis, feature engineering, categorical encoding, time-based validation, model training, and evaluation.

Three machine learning algorithms—Linear Regression, Random Forest, and XGBoost—were developed and compared.

Among the three models, **Random Forest Regressor achieved the best overall performance with an MAE of 88.32, RMSE of 196.48, and R² Score of 0.714.**

The final model was used to generate future food demand predictions. These predictions can be used to support food preparation, inventory planning, food-waste reduction, and prevention of stock shortages.

Overall, the project demonstrates the practical application of machine learning to a real-world demand forecasting problem and shows how predictive analytics can support operational decision-making in the food-service industry.